Samarth Dalela

Software Developer | Full-Stack (MERN / PERN) & Microservices

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A-153, Lajpat Nagar-1, New Delhi
B.Tech (CSE) | DOB: 12/05/2002

PROFESSIONAL SUMMARY

Software Developer with **2 years** of experience at NIELIT HQ, Delhi, plus part-time consulting at M37Labs on AI-based PPT generation, building full-stack web applications, microservices with **Apache Kafka**, a PostgreSQL-based ticketing system, a CMS, an AI chatbot and payment-enabled conference websites. Experienced in **server and hosting management** (Nginx, Windows Server, NSSM, PM2, Vercel), **CI/CD pipelines**, **MySQL and Redis optimization**, **API load testing** (k6, Postman), and securing applications against the OWASP Top 10. Seeking a challenging role to apply strong coding and problem-solving skills to innovative projects.

TECHNICAL SKILLS

Frontend	React.js, HTML5, CSS3, Bootstrap, EJS	**Backend**	Node.js, Express.js, PHP, REST APIs
Databases	PostgreSQL, MongoDB, MySQL (query & index optimization), Redis (caching)	**Architecture**	Microservices, Apache Kafka (message broker, event streaming, buffering)
DevOps & Hosting	CI/CD pipelines, Vercel, Nginx, NSSM, PM2, Windows Server, SSL/TLS	**Other**	Chrome Extensions, Moodle LMS, WordPress, Nodemailer / SMTP, Python web scraping
Security	OWASP Top 10 audits, JWT, input validation, secure headers	**Testing**	API testing (Postman), load & stress testing (k6)
Languages & Tools	JavaScript, Python, C++, C# (learning), ASP.NET, Git, GitHub		

WORK EXPERIENCE

Software Developer — National Institute of Electronics and Information Technology (NIELIT HQ), New Delhi **2 Years**

1. NIELIT Main Website Chatbot

Node.js · Express · React

- Designed and developed an AI-powered chatbot integrated into the NIELIT main website
- Built using Node.js and Express.js backend with REST API architecture
- Implemented natural language understanding to handle user queries related to NIELIT courses, admissions, and services
- Integrated chatbot widget into the frontend using React.js for seamless user interaction

2. Content Management System (CMS) Development

MERN · PM2 · Windows Server

- Developed a custom CMS for managing website content, news, events, and announcements for NIELIT main website
- Built with MERN stack (MongoDB, Express.js, React.js, Node.js) with role-based access control
- Implemented CRUD operations, media management, and dynamic page generation
- Designed admin dashboard for non-technical staff to manage content without coding knowledge
- Deployed and hosted the CMS on Windows Server 2025 — configured environment variables, Node.js runtime, and PM2 process manager for production stability and auto-restart on failure
- Performed OWASP Top 10 security audit on the CMS — identified and resolved vulnerabilities including SQL Injection, XSS, CSRF, Broken Access Control, Security Misconfiguration, and Sensitive Data Exposure; applied input sanitization, JWT-based authentication hardening, rate limiting, and secure HTTP response headers as remediation

3. Moodle LMS (Learning Management System) Development

Moodle · PHP

- Configured, customized, and deployed Moodle LMS for NIELIT's online training and e-learning programs
- Developed custom Moodle plugins and themes to match institutional branding
- Managed course creation, student enrollment, assessment modules, and reporting features
- Integrated Moodle with NIELIT's internal systems for seamless data flow

4. UPWIECON Conference Website (PHP)

PHP · Nginx · EaseBuzz

- Developed the official conference website for UPWIECON 2025 using PHP
- Developed the official conference website for UPWIECON 2026 using PHP
- Built dynamic pages for paper submission, schedule, speakers, registration, and committee listings
- Implemented PHP-based backend for form handling, email notifications, and database management
- Integrated and configured EaseBuzz payment gateway for online registration fee collection — implemented payment initiation, success/failure callback handling, and transaction status verification via EaseBuzz PHP SDK
- Deployed and hosted both UPWIECON websites on Nginx web server — configured virtual hosts, PHP-FPM, server blocks, SSL/TLS certificates, and URL rewriting for clean URLs and HTTPS enforcement

- Performed OWASP Top 10 security audit across UPWIECON PHP websites — identified and resolved vulnerabilities including SQL Injection, XSS (Cross-Site Scripting), CSRF, Broken Authentication, Security Misconfiguration, Sensitive Data Exposure, and Insecure Direct Object References (IDOR); implemented input validation, prepared statements, output encoding, and secure HTTP headers as remediation
- Ensured mobile-responsive design and cross-browser compatibility

5. NICEDT Conference Website (WordPress)

WordPress · BillDesk

- Developed and maintained the NICEDT (National International Conference on Emerging and Disruptive Technologies) website using WordPress
- Customized WordPress themes and plugins to meet conference requirements
- Managed conference content including paper submission guidelines, important dates, keynote speakers, and program schedule
- Integrated and configured BillDesk payment gateway for secure online conference registration fee collection — set up API credentials, webhook handlers, and transaction verification for seamless payment processing
- Conducted OWASP Top 10 security audit on the WordPress site — remediated vulnerabilities including XSS, CSRF, broken access control, sensitive data exposure, and security misconfigurations; hardened WordPress configuration, restricted file permissions, and enforced secure HTTP headers
- Configured SEO, security plugins, and performance optimization for production deployment

6. Microservice Architecture with Apache Kafka

Node.js · Kafka · Microservices

- Built a microservice architecture for NIELIT applications, splitting services into independently deployable units
- Used Apache Kafka as the message broker for asynchronous, event-driven communication between services
- Used Kafka topics as a message buffer to decouple producers and consumers, absorb traffic spikes and prevent message loss when a service is down

7. Server, Hosting & Deployment Management

Windows Server · NSSM · Nginx · Vercel · CI/CD

- Managed NIELIT application servers and hosting: Windows Server 2025, Nginx with PHP-FPM, virtual hosts, SSL/TLS and HTTPS
- Implemented NSSM to run Node.js applications as Windows services, so they start on boot and restart automatically on failure
- Set up CI/CD pipelines for automated build and deployment, reducing manual release steps
- Deployed frontend applications on Vercel with environment configuration and production builds

8. Database & Caching Optimization

MySQL · Redis

- Optimized MySQL performance through indexing, query tuning and analysis of slow queries
- Implemented and optimized Redis caching for frequently accessed data, with expiry (TTL) policies to cut database load and response times

9. Ticketing System

PostgreSQL · Express · React · Node.js

- Developed a ticketing system with ticket creation, assignment, priority, status tracking and resolution workflow
- Designed the PostgreSQL schema, built REST APIs in Node.js/Express and a React dashboard with role-based access

10. Auto-Translation Chrome Extension

JavaScript · Chrome Extension

- Developed a Chrome extension for automatic translation, integrated with the NIELIT CMS to speed up multilingual content publishing

11. NIC SMTP Relay Email Integration

Node.js · Nodemailer · SMTP

- Integrated NIC's government SMTP mail and OTP relays into NIELIT applications using Nodemailer with STARTTLS and authenticated login
- Diagnosed authentication and sender-whitelisting issues with NIC Relay Support and verified delivery end to end

12. Tender Scraper & CMS Auto-Uploader

Python · BeautifulSoup · REST API

- Built a Python tool that scrapes tender notices and documents from the NIELIT website and uploads them to the CMS through its API

13. API Testing & Load Testing

k6 · Postman

- Tested REST APIs with Postman: built request collections, environments and test scripts to validate responses, status codes and authentication
- Performed load and stress testing on APIs with k6, simulating concurrent virtual users to measure response times, throughput and error rates
- Used test results to find bottlenecks and verify API stability under load before production releases

14. BCAS Website Maintenance (.NET)

ASP.NET · C# · HTML · CSS · JavaScript

- Maintained the BCAS website built on .NET, handling frontend updates, UI fixes and content changes
- Worked with the C# codebase while learning C# and ASP.NET to support backend changes

Also: Python-based data analysis projects; support for NIELIT digital learning infrastructure and training delivery.

Consultant (Part-time) — M37Labs **2 Months**

AI-Powered Automated PPT Generation Software

AI · Automation

- Worked as a part-time consultant on software that automatically generates PowerPoint presentations using AI
- Contributed to turning user input into structured, ready-to-use slide decks, reducing manual presentation effort

EDUCATION

Qualification	Institution	Board / University	Year	Score
B.Tech (CSE, Hons)	Maharana Pratap College of Technology, Gwalior	Rajiv Gandhi Proudyogiki Vishwavidyalaya	2024	87.9%
Class XII	Kendriya Vidyalaya No.1, Gwalior	CBSE	2020	87%
Class X	Kendriya Vidyalaya No.1, Gwalior	CBSE	2018	88%

INTERNSHIPS & CERTIFICATIONS

Internships

- MERN — NIELIT
- Data Science — NIELIT
- AICTE MERN — Shripriti Educational & IT Hub
- Web Development — AD Infocom System, Pune
- Web Development — BreachCurls, Gwalior
- AICTE Cyber Security — IBM (Edunet Foundation)

Certifications

- MERN Full Stack — NIELIT
- Data Analysis using Python — NIELIT
- MERN Full Stack — Udemy
- C++ — Skill India (2020–21)
- Python — Udemy
- HTML5 — Udemy

ACADEMIC PROJECTS

AskMe Web App (MERN, major project) · Virtual Personal Assistant (Python, NLP) · Desktop Manager (C++) · Resume Builder & QR Code Generator (AD Infocom) · Educational websites for Shripriti & Coding Guruji · RedCliff Hotel & Gym websites · Rock Paper Scissors (Python) · RGPV News Website (HTML5)

ACHIEVEMENTS

- Secured **First Position** in 3rd year of B.Tech (2023)
- Awarded **B.Tech Honours Degree** (2024)

PERSONAL DETAILS

Father's Name	Avinash Dalela	**Mother's Name**	Preeti Dalela
Date of Birth	12/05/2002	**Gender**	Male
Marital Status	Unmarried	**Languages**	English, Hindi
Address	A-153, Lajpat Nagar-1, New Delhi	**Interests**	Coding, Animation, Photography, Painting, Reading & Writing

DECLARATION

I hereby declare that all the information provided above is true and correct to the best of my knowledge and belief.

Place: New Delhi (Samarth Dalela)

Date: _____